
Mobile and Ubiquitous Computing

Mobile Applications Design



FCUL, Departamento de Informática

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The fundamental problem

- For **who** do we design for?
- For **what**, **when** and **where** will it be used?
- What tools and methods and ... do we have to design with?
- What are we expected to design?

... in life, around the world ...



everybody, everywhere, everytime

everybody independently of culture, literacy, age, ...

... multiple (de)form(ed) factors

What do we have to work with:

- 1) very strange form factors yet requiring continuity (ubiquity)
- 2) From powerful versatile hardware (smartphones, foldables, tablets), to very specific, limited ones (smartwatches, fitness bands, GPS and sports devices)



... and today's form factors

- **Phones** from ~6" to 6.9", plus foldables and flip phones: one device, several screen states
- **Tablets and laptops** continue the same task: handoff, sync, cross-device continuity
- **Wearables** (watches, rings, bands): glanceable, seconds-long interactions
- **Smart glasses and XR headsets** (e.g., Ray-Ban Meta, Apple Vision Pro)
- **Car and home** (CarPlay, Android Auto, smart speakers): eyes-free, voice-first

Design implications: adaptive layouts and state continuity when folding, rotating or switching devices

Phone

Foldable

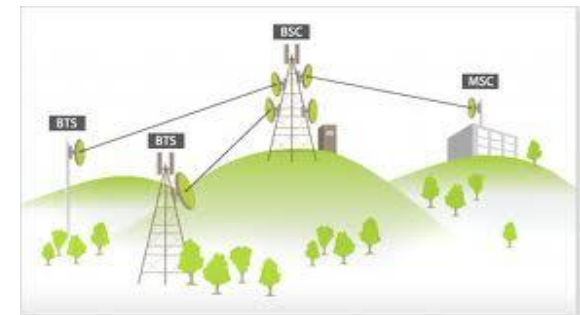
Watch

Glasses / XR

Car & voice

... out of the ordinary tech ...

Networking, sensors,



... Extraordinary interfaces

truthfully **multimodal** user interactions

vision, hearing, touch, sound and voice, micro and macro gestures



... and AI joins the multimodal mix

- **Conversational assistants** (LLM-based): natural language as a general input modality, by voice or text
- **Camera as input**: visual search, live translation, text recognition
- **On-device AI**: summarising, writing, image editing, live captions, screen descriptions
- **Design implications**
 - set expectations: what the system can and cannot do
 - make AI output visible, editable and undoable; signal uncertainty
 - privacy: what leaves the device, explicit consent

See: Amershi et al., "Guidelines for Human-AI Interaction", CHI 2019 · Google PAIR, People + AI Guidebook

Enrichment/impoverishment of application & interaction

Contradictory forces:

- 1) We have the tech and the context to develop extraordinary applications
- 2) We have limitations of size, battery, context interruption and cluttering, computational power, intermittent network access...



Can we use standard guidelines?

Do usability basics apply?

What does user experience mean in this context?

THE OUT OF THE BOX...

Maintain good practices

Guidelines apply!



Utility



Emotion

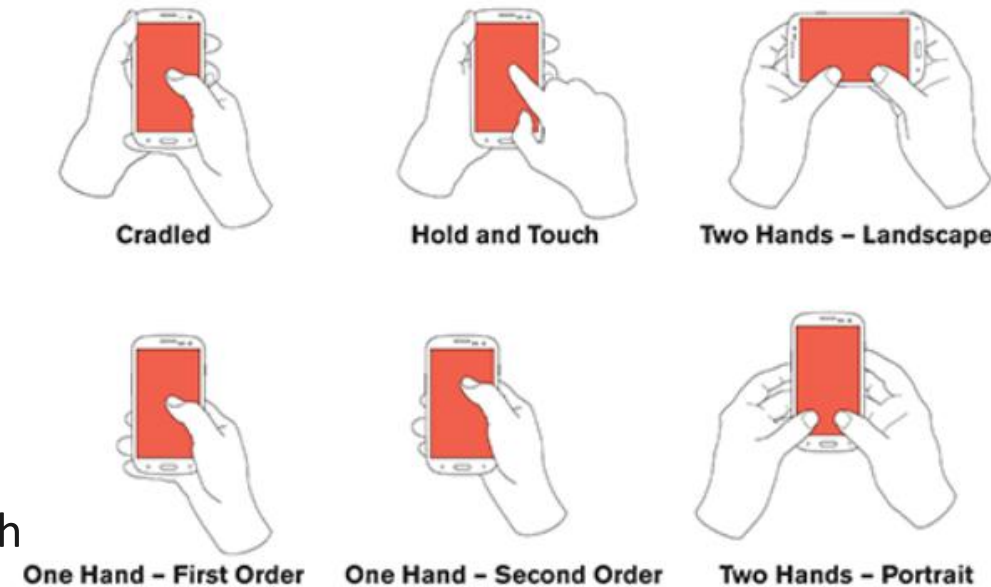
coherence, shortcuts, feedback, closure,
avoid errors, undo, control, memory, ...

Some Guidelines ...

1. Device diversity is human diversity
2. People touch/look at the center of the screen
3. Fingers get in the way
4. People use different devices in different ways
5. One, two, three for better mobile design
6. Phones are not flat
7. Thumbs, gestures and big screens
8. Touch targets: the numbers

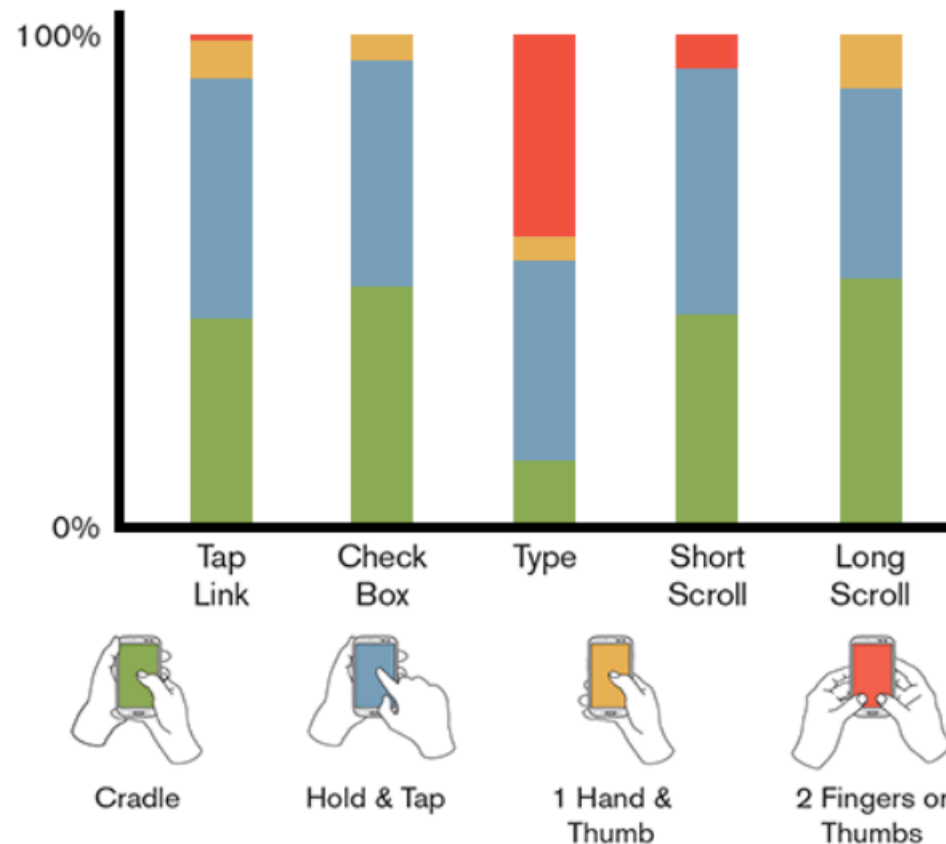
1. Device diversity is human diversity

- People hold phones in multiple ways, depending on their device, their needs, and their context
- They change their methods of grasping their phone without realizing it
- 75% of users **touch the screen** only with one thumb
- Fewer than 50% of users **hold** their phone with one hand
- 36% of users **cradle** their phone, using their second hand for both greater reach and stability
- 10% of users **hold their phone** in one hand and **tap** with a finger of the other hand.



1. Device diversity is human diversity

- How people change their grip for specific types of interactions, depending on their task and context



1. Device diversity is human diversity

- Design for
 - every user and every type and size of the phone
 - the variable ways in which people work with their devices, not just one-way
- Don't assume everyone has your phone or uses it the same way you do

Source: S. Hooper, "Design for Fingers, Touch, and People" (series), UXmatters, 2017

2. People touch the center of the screen

- Users prefer to touch the center of the screen and will do so whenever you give them that choice
- Place key actions in the middle half to two-thirds of the screen
- Place secondary options and buttons that provide paths to secondary screens along the top and bottom of the screen.



Source: S. Hooper, "Design for Fingers, Touch, and People" (series), UXmatters, 2017

3. Fingers get in the way

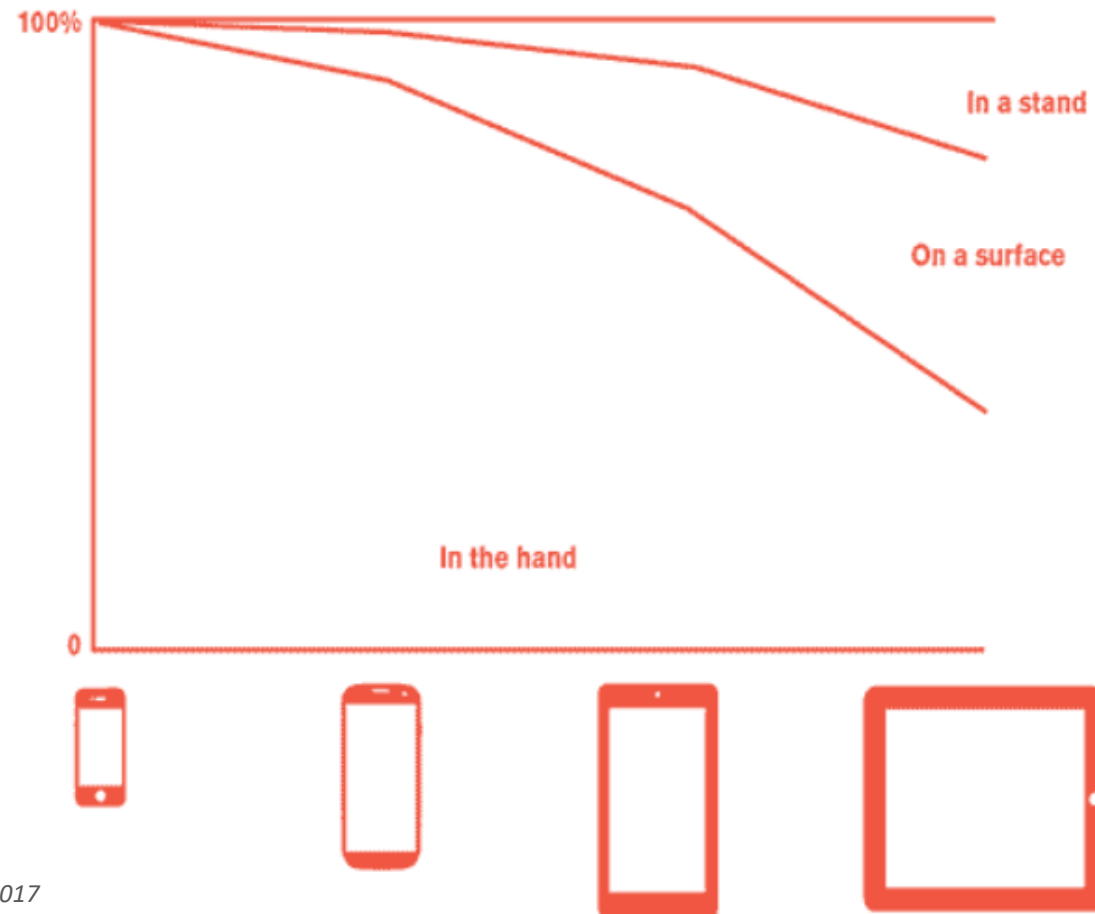
- Make sure **people's fingers and thumbs** don't obscure content, so they can see what they're tapping
- Make sure **selectable items** are large enough that they can clearly indicate a successful tap
- **Place functions or any content that changes state** to allow users to see the result
- Ensure there is **plenty of whitespace within the content**, as well as reasonable margins, so users can feel confident about scrolling or gesturing in an area that's separate from any selectable content



Source: S. Hooper, "Design for Fingers, Touch, and People" (series), UXmatters, 2017

4. People use different devices in different ways

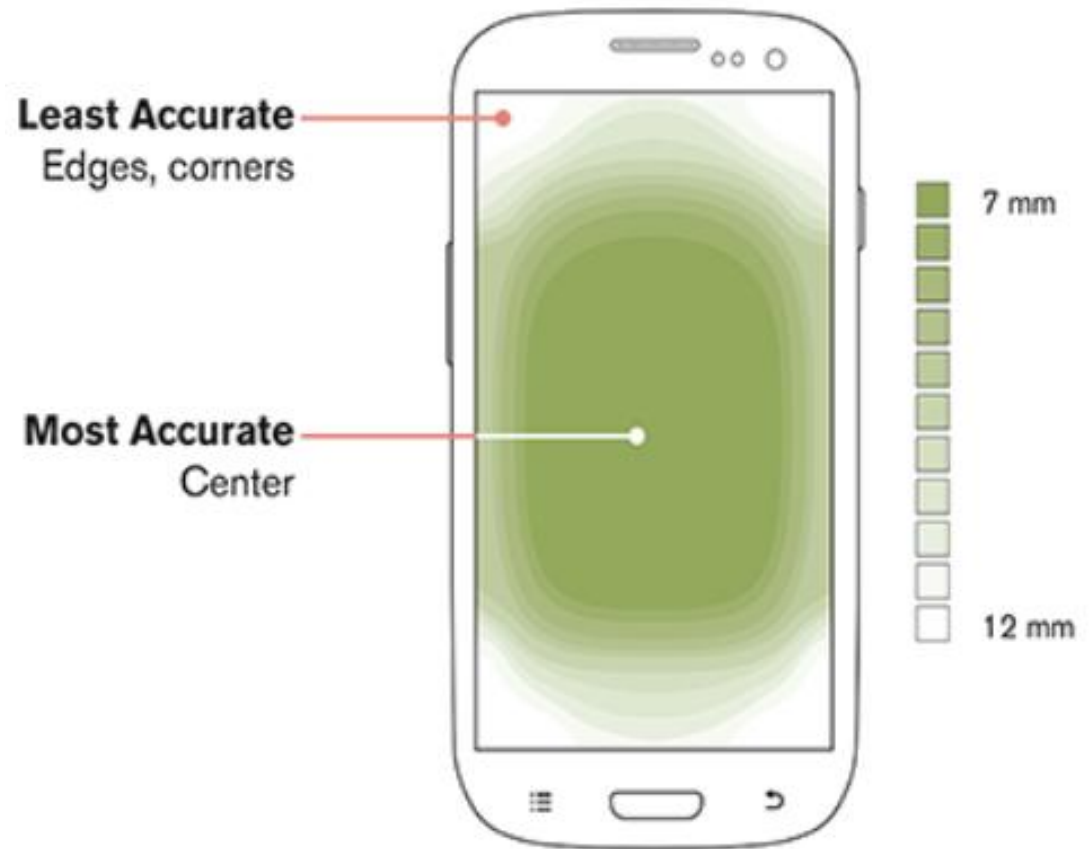
- **Support all input types**—especially if you are building an app for tablets and mobile phones
- **Gather data** on how your users work in their actual environment
- For **optimal legibility**, adjust the sizes of type, icons, text boxes, checkboxes, and buttons to accommodate the typical distance of the screen from the user's eyes for a particular device class



Source: S. Hooper, "Design for Fingers, Touch, and People" (series), UXmatters, 2017

5. One, two, three for better mobile design

- People's touch interactions are faster, more confident, and more accurate as they approach the center of the screen

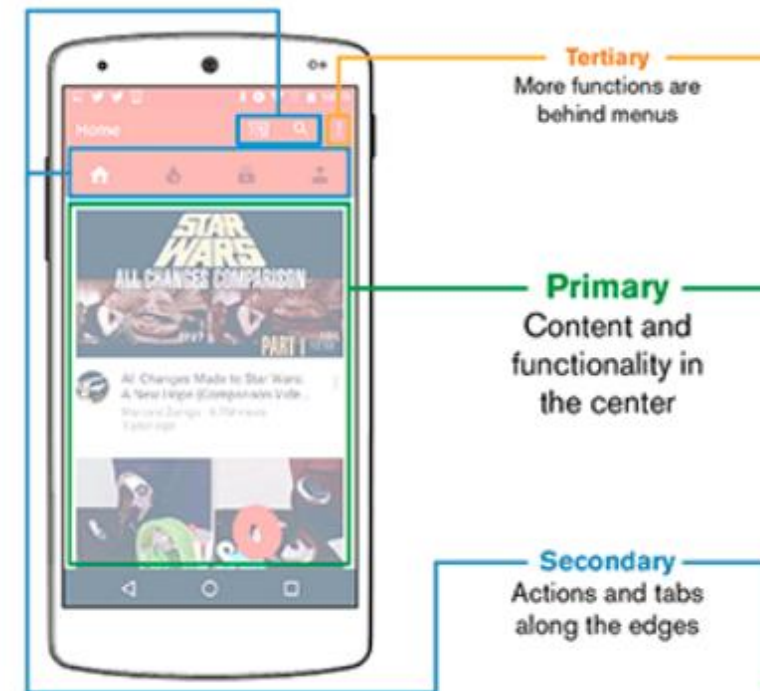


Source: S. Hooper, "Design for Fingers, Touch, and People" (series), UXmatters, 2017

5. One, two, three for better mobile design

Information design framework

- Place **secondary actions** along the top and bottom edges of the screen
- Tabs along the top or bottom edge of the content area let users switch views or sections. Action buttons let users compose content or search
- Hide **tertiary functions** on menus that users can open by tapping a target in one of the corners of the screen.

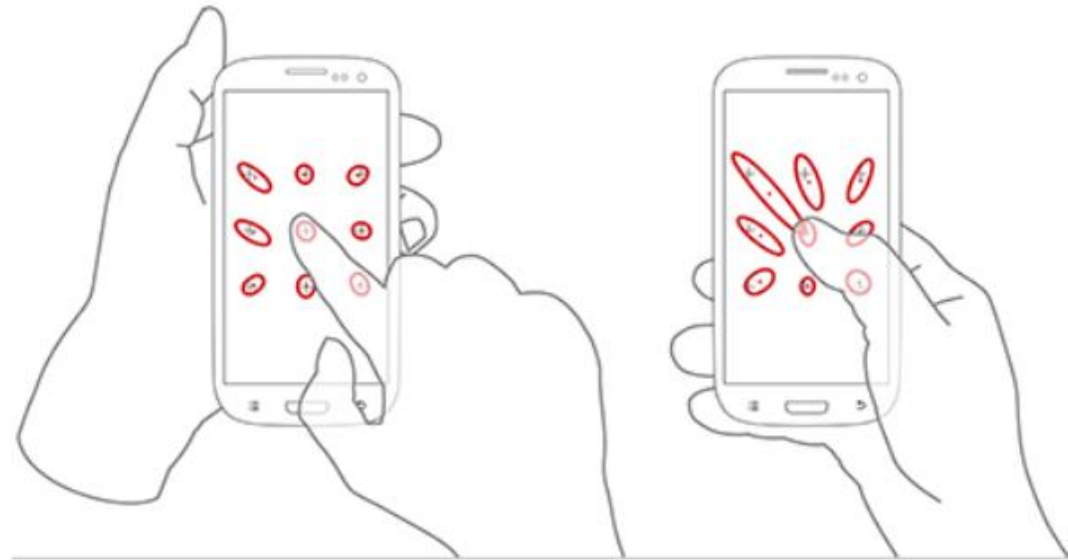


6. Phones are not flat

- People shift the way they hold and touch their devices depending on their context.

Some examples:

- opening a door
- carrying groceries, baby, ...
- walking down the street
- walking in difficult terrain



differences in accuracy for nine touch targets
when a user is holding the phone in two hands
vs one hand, while carrying a bag, on the right

Source: S. Hooper, "Design for Fingers, Touch, and People" (series), UXmatters, 2017

6. Phones are not flat

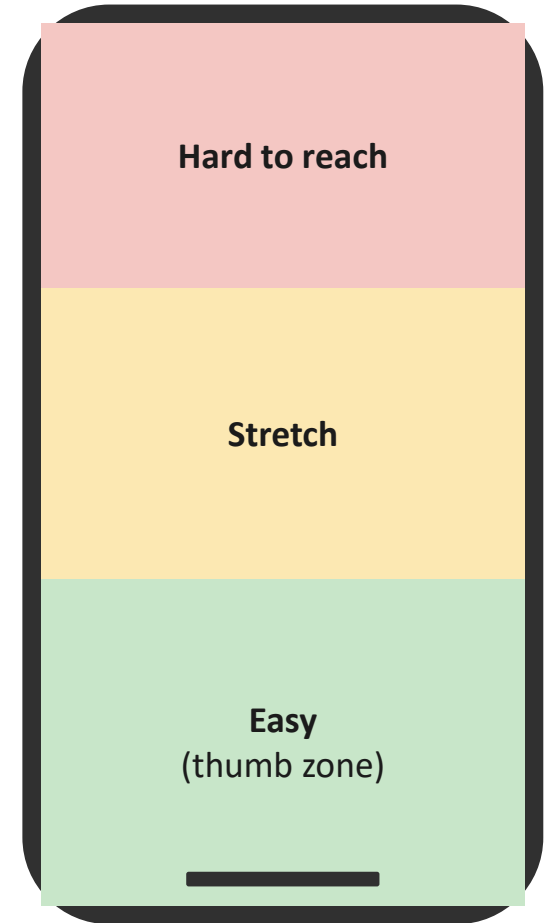
- Avoid getting too close to the boundaries of any touch-target size requirements
- Plan for different contexts, and
 - try out your designs in difficult environments to make sure they work well in the real world
- If you need to place items right against the edges of the screen or use edge gestures
 - provide some padding so they'll work even with cases and curved screens

Source: S. Hooper, "Design for Fingers, Touch, and People" (series), UXmatters, 2017

7. Thumbs, gestures and big screens

- Phones got taller ($\approx 6.1''$ – $6.9''$): the top is out of thumb reach in one-handed use
- The OS now owns the edges: home swipe on the bottom edge, back swipe on the side edges
- Put **primary navigation and frequent actions at the bottom** (tab bar, navigation bar)
- Put rare actions at the top; keep destructive actions away from frequent ones and ask for confirmation
- No custom gestures or tappable items right on the edges

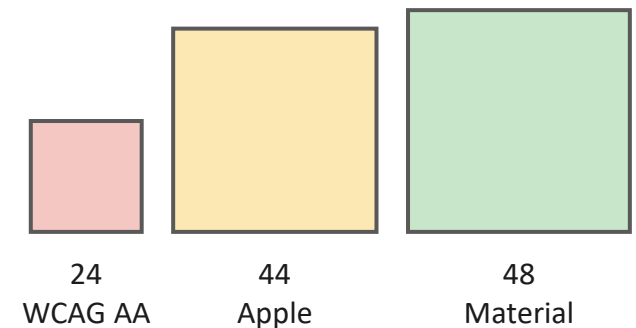
Revisits guideline 5



Simplified one-handed reach; bar = system gesture area

8. Touch targets: the numbers

- **Apple HIG**: at least 44×44 pt (7 mm)
- **Material Design**: at least 48×48 dp (≈ 7 - 10 mm)
- **WCAG 2.2 - Web Content Accessibility Guidelines (2023)**, SC 2.5.8 (AA): at least 24×24 CSS px, or enough spacing around it
- WCAG SC 2.5.5 (AAA): 44×44 CSS px
- The icon can be small; the **hit area** must not be
- Accuracy drops at edges and corners: give targets there more room (guidelines 5 and 6)



Drawn to the same scale

pt – points; dp - density-independent pixel

CSS px é o "píxel de referência" da web

Unit that does not depend on the screen resolution

Accessibility is no longer optional

- **European Accessibility Act** (Directive (EU) 2019/882), applicable since 28 June 2025: covers apps for e-commerce, banking, transport, e-books, communications
- Reference standard EN 301 549 → WCAG 2.1 AA (aim for WCAG 2.2)
- In practice
 - labels for every control (VoiceOver, TalkBack)
 - text scaling (Dynamic Type / font scale) without broken layouts
 - contrast $\geq 4.5:1$; never colour alone to convey meaning
 - respect “reduce motion”; captions; both orientations
- Test with assistive technologies and with disabled users

28 June 2025

EAA applies to new products and services in the EU

Follow the platform

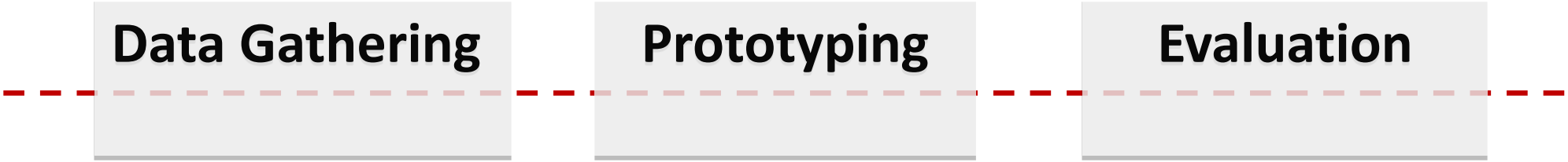
- Platforms shape user expectations: follow their conventions before inventing new ones
- **Native components** come with accessibility, gestures and consistency for free
- Respect system settings: dark mode, dynamic colour, text size, language
- **Adaptive layouts** for phones, foldables and tablets (size classes / window size classes)

iOS / iPadOS

Human Interface Guidelines
Liquid Glass (iOS 26, 2025)
developer.apple.com/design

Android

Material Design 3
M3 Expressive (2025)
m3.material.io



Data Gathering

Prototyping

Evaluation

How to do it?

Where to do it?

What to get out of it?

UNDERSTANDING FOR DESIGN

Data Gathering

Prototyping

Evaluation

Early problems regarding

Data Gathering

Almost impossible to list every single requirement

Prototyping

Needed to evolve from just using the classic “sketch cards” given the unique characteristics of mobile devices

Evaluation

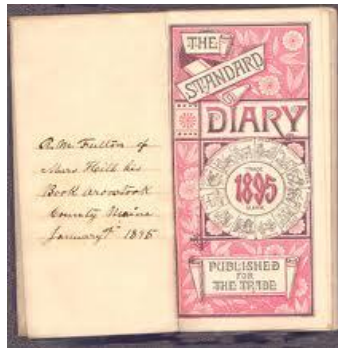
There was an absence of evaluation methods for mobile application and this caused this crucial step to be neglected many times

Based on: M. de Sá & L. Carriço, “Lessons from early stages design of mobile applications”, MobileHCI 2008

Gathering data techniques



observation, inquiring, focus groups, lab tests, diaries and ...



Everything applies + Context

Understand data and context transitions

Scenario Generation Framework

Contextual Scenarios

Scenarios composed by a group of variables

Scenario Transitions

Transitions between contextual scenarios

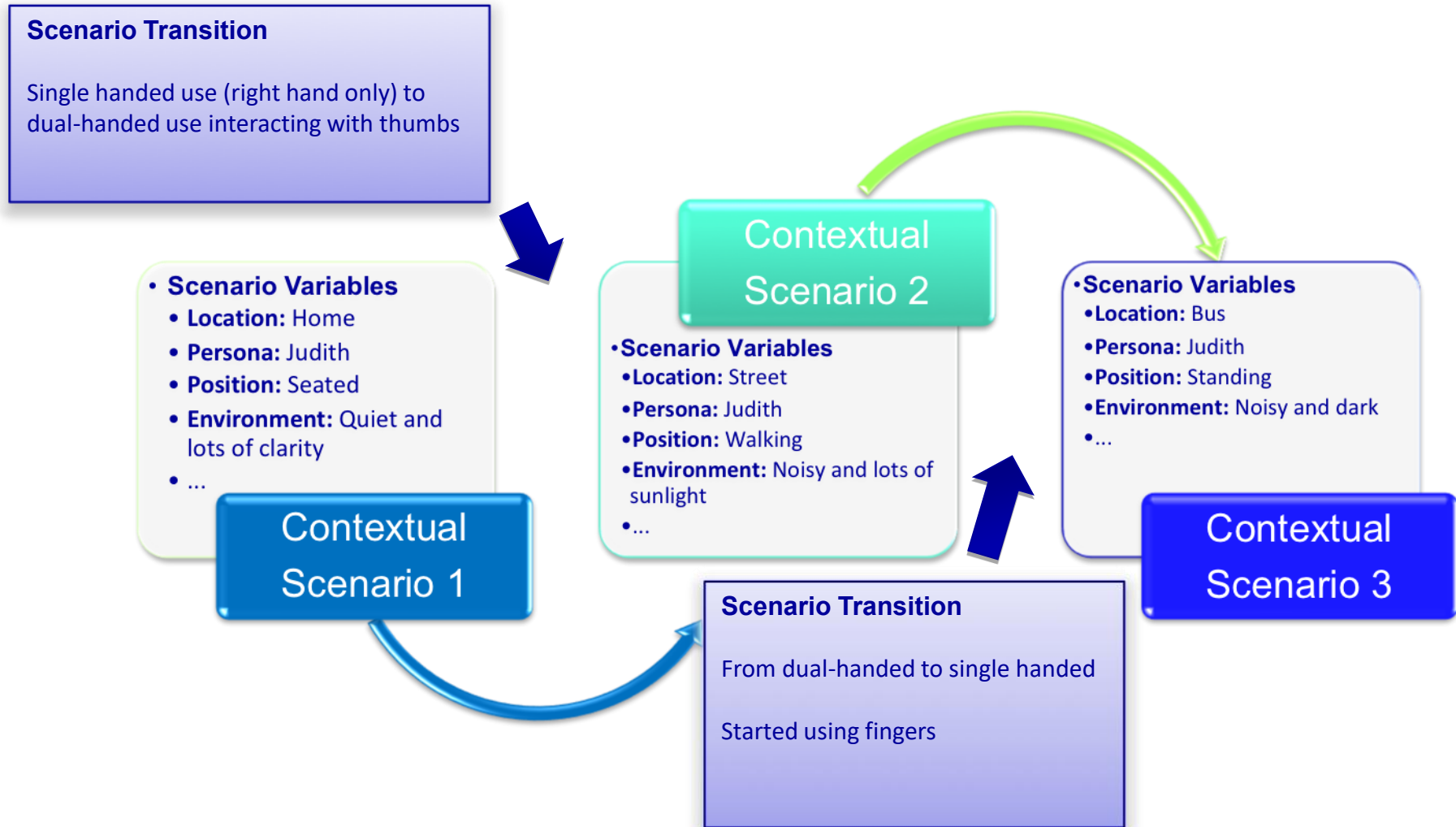
Scenario variables

Details that compose each contextual scenario.

- Locations and Settings
- Movement and Posture
- Workloads, Distractions and Activities
- Devices and Usages
- **User Personas**

Based on: M. de Sá & L. Carriço, "Defining scenarios for mobile design and evaluation", CHI 2008 Extended Abstracts

Understand Data and Context Transitions



-
- What to aim for?
 - How accurate?
 - What for?

DESIGNING & PROTOTYPING

Designing

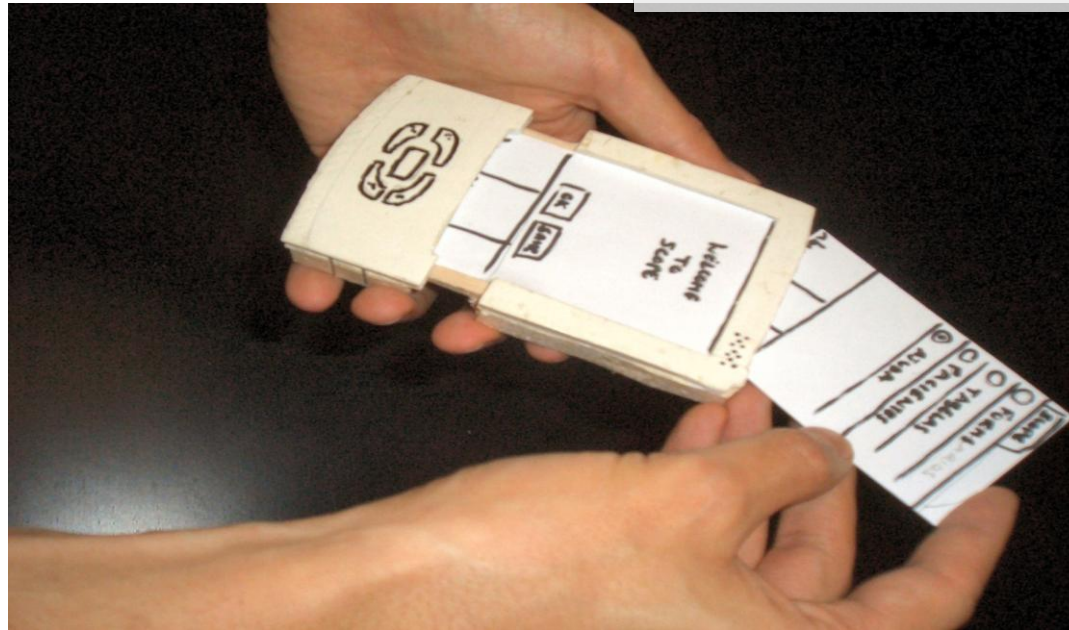
When designing, besides what you have understood, also consider:
fun, battery and the eco stuff, diverse interaction modes, including one-hand interaction, branding (most of them should also emerge from understanding).



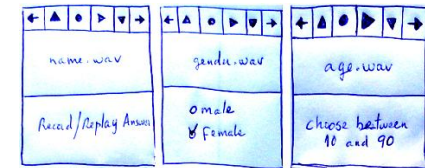
Prototypes



Card Sketch
Hand Drawn Card

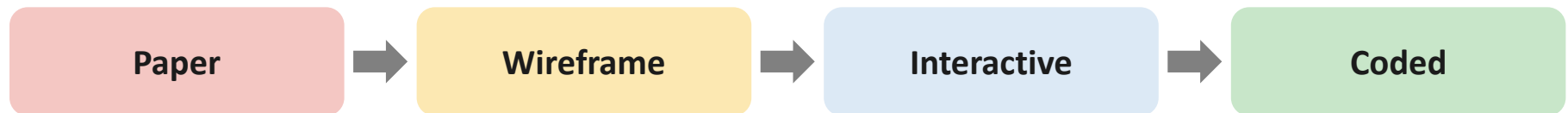


Rigid Frame
Cards can be easily inserted and removed



Prototyping today

- Still **start low-fi**: paper, sketches, cards; cheap to change
- **Interactive prototypes** in Figma (run them on a phone with the Figma mobile app); ProtoPie for sensors and complex interactions
- Prototype with the platform **UI kits** (Apple, Material 3)
- **AI-assisted** screen and code generation: fast exploration; check consistency, accessibility and platform conventions
- Always test **on a real device, in hand, in context**, not only on a monitor



With whom?

With what?

Where?

EVALUATING

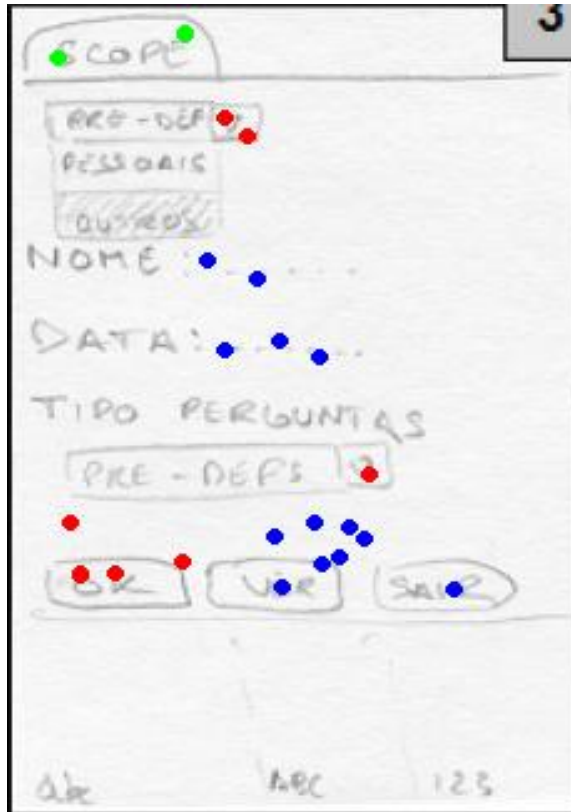
Evaluation of mobile apps

- Physical context – where it will be used
- Technological context – how it will be used
e.g. network access and bandwidth
- Social context – by whom it will be used

Evaluate with whomever is available (almost)



Evaluation with all prototypes



On Low-Fi

- Use different colors for different tasks
- Use screen tagging for contextual card sorting
- Use scenarios with transitions
- Allow participatory design and in-situ sketching

Evaluation with all prototypes :)



Evaluate here, there and everywhere



Laboratory
Living simulated settings
Target worlds



Evaluation with whomever, wherever

- Use markets/stores and log, log, log
- Social networks...



App Store



Google play

Evaluation in the wild, today

A mobile app is evaluated using real users in their own context in five ways:

- **Beta channels:** TestFlight (iOS), Google Play testing tracks
- **Product analytics** and event logs (e.g., Firebase): funnels, retention, drop-off
- **Compare versions:** A/B tests and gradual rollouts, starting with a small percentage of users
- **Crash and performance reports** (e.g., Crashlytics)
- **Store ratings and reviews** as qualitative data
- **Remote usability tests** (moderated or not) with screen and video recording
- Logs show **what**, not **why**: combine with interviews and observation
 - Ethics: informed consent, data minimization, GDPR

Standard instruments

- **Questionnaires:** SUS (System Usability Scale), UEQ (User Experience Questionnaire), SEQ (after each task), NASA-TLX (workload)
- **Heuristics:** Nielsen's 10, complemented by mobile-specific sets (e.g., Bertini, Gabrielli & Kimani, AVI 2006)
- **Accessibility checks:** Accessibility Inspector (Xcode), Accessibility Scanner (Android), plus manual testing with VoiceOver / TalkBack
- **Metrics:** task success, time on task, errors, measured in context and on real devices

Evaluation with Users

	Quick & Dirty	Conceptual Model Extraction	Direct Observation	Interviews	Questionnaires
Conducted by	End-users	End-users	End-users	End-users	End-users
Equipment	Low-Fi prototype	Interface sketch	Interactive prototype	Interactive prototype	Interactive prototype & questionnaire
Results	Anecdotal	Qualitative	Qualitative	Qualitative	Quantitative & Qualitative
Where	Informal setting	Controlled setting	Controlled setting	Controlled setting	Anywhere

Evaluation without users

	Heuristic	No people
Conducted by	Usability experts	Theoretical model
Equipment	Interactive prototype	Formal specification of the interaction
Results	Qualitative & Quantitative	Quantitative
Where	Informal setting	Anywhere